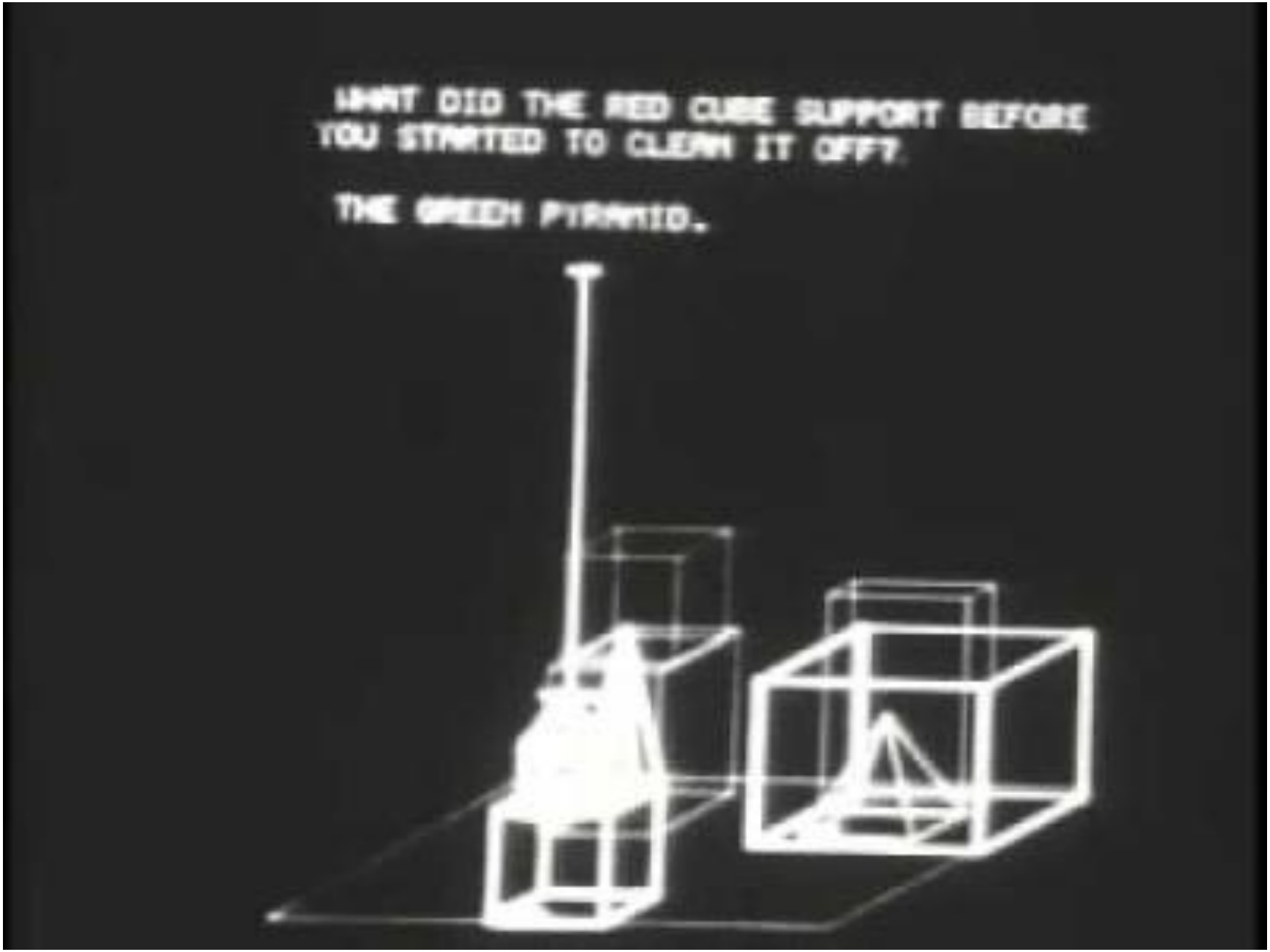
CS 380: ARTIFICIAL INTELLIGENCE

Lecture 9: Natural Language Processing

Natural Language Processing

- Let's start with some classic examples:
  - Eliza (Weizenbaum, 1966)
    - Sample implementation: http://www.masswerk.at/elizabot/
  - SHRDLU (Winograd, 1970)
    - http://hci.stanford.edu/~winograd/shrdlu/
    - https://www.youtube.com/watch?v=QAJz4YKUwqw
    - How did SHRDLU get its name? http://hci.stanford.edu/~winograd/shrdlu/name.html


WHAT DID THE RED CUBE SUPPORT BEFORE YOU STARTED TO CLEAN IT OFF?
THE GREEN PYRAMID.

Natural Language Processing

- SHRDLU is quite impressive for late 1960s!
- So, is NLP solved nowadays? (after ~50 years?)
  - Not at all!
- But SHRDLU worked pretty well — can't we just apply the algorithms in SHRDLU to other domains?
  - SHRDLU worked on a "micro-world", which masks most of the hard problems in NLP (ambiguity, metaphor, non-compositionality, etc.)
  - Efforts to scale SHRDLU to larger domains have consistently failed (e.g., CyC)

Natural Language Processing

- There are many possible perspectives for an analysis of language…
  - Linguistics: e.g., general & specific aspects of language
  - Psychology: e.g., developmental learning
  - Computer Science: e.g., grammar & parsing
  - Sociology: e.g., language across cultures
  - Etc.
- Here we focus on NLP from an **agent** perspective: an agent/human communicating with an agent/human

Communication

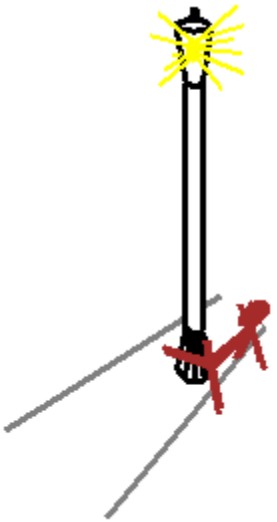
- "Classical" view (pre-1953)
  - Language consists of sentences that are true/false (logic)
- "Modern" view (post-1953)
  - Language is a form of action

Wittgenstein(1953) Philosophical Investigations

Austin (1962) How to Do Things with Words

Searle (1969) Speech Acts

- Why?
  - To change the actions of other agents



Speech Acts



- Speech acts achieve the speaker´s goals:
  - Inform "There's a pit in front of you"
  - Query "Can you see the gold?
  - Command "Pick it up"
  - Promise "I'll share the gold with you"
  - Acknowledge "OK"
- Speech act planning requires knowledge of
  - Situation
  - Semantic and syntactic conventions
  - Hearer's goals, knowledge base, and rationality

Stages of Communication (informing)

- Intention — S wants to inform H that P
- Generation — S selects words W to express P in context C
- Synthesis — S utters words W

- Perception — H perceives W' in context C'
- Analysis — H infers possible meanings $P_1, \ldots P_n$
- Disambiguation — H infers intended meaning P_i
- Incorporation — H incorporates P_i into KB

How could this go wrong?

Stages of Communication (informing)

- Intention — S wants to inform H that P
- Generation — S selects words W to express P in context C
- Synthesis — S utters words W

- Perception — H perceives W' in context C'
- Analysis — H infers possible meanings $P_1, \ldots P_n$
- Disambiguation — H infers intended meaning P_i
- Incorporation — H incorporates P_i into KB

How could this go wrong?

- Insincerity (S doesn't believe P)
- Speech wreck ignition failure (speech recognition failure ☺)
- Ambiguous utterance
- Differing understanding of current context (C != C')

Stages of Communication (informing)

- Intention — S wants to inform H that P
- Generation — S selects words W to express P in context C
- Synthesis — S utters words W

- Perception — H perceives W' in context C'
- Analysis — H infers possible meanings $P_1, \ldots P_n$
- Disambiguation — H infers intended meaning P_i
- Incorporation — H incorporates P_i into KB

Ambiguity examples from NYT Crossword:

They hang around in kitchens = ?

Bare-bones outfit = ?

Device for on-line conversation? = ?

Grammar

- Vervet monkeys, antelopes, etc. use isolated symbols for sentences
  - Restricted set of communicable propositions, no generative capacity

Grammar

- Vervet monkeys, antelopes, etc. use isolated sym[illegible]sentences



Vervet monkeys are known to elicit **predator-specific alarm calls**. Three well-documented vervet monkey alarm calls are those for leopard, martial eagle and python. **Leopard alarm calls are short tonal calls** produced in a series of inhalations and exhalations. **Eagle alarm calls are low pitched grunts** while python alarm calls are high pitched "chutters". Different alarm calls seem to evoke different responses to individuals that heard the alarm calls. However, **the first reaction of a vervet monkey upon hearing an alarm call is to look at the direction of the caller**. Looking at the direction of the caller gives them clues as to why the alarm calls were made and also where the caller is facing reveals the direction of the approaching predator. *[from primatology.net]*

Grammar

- Vervet monkeys, antelopes, etc. use isolated symbols for sentences
  - Restricted set of communicable propositions, no generative capacity

(Chomsky (1957): Syntactic Structures)

- Grammar specifies the compositional structure of complex messages. E.g., speech (linear), text (linear), music (two-dimensional)
- A formal language is a set of strings of terminal symbols
- Each string in the language can be analyzed/generated by the grammar
- The grammar is a set of rewrite rules, e.g.,

  S → NP VP

  Article → the | a | an | …

  Backus-Naur Form (BNF)

  Here S is the sentence symbol, NP and VP are nonterminals

Grammar Types

- Regular: nonterminal → terminal[nonterminal]

  $S \rightarrow aS$

  $S \rightarrow \Lambda$ (Λ = uppercase lambda = empty string)
- Context-free: nonterminal → anything

  $S \rightarrow aSb$
- Context-sensitive: more nonterminals

  $ASB \rightarrow AAaBB$
- Recursively enumerable: no constraints
- Natural languages probably context-free, parsable in real time!

Wumpus Lexicon

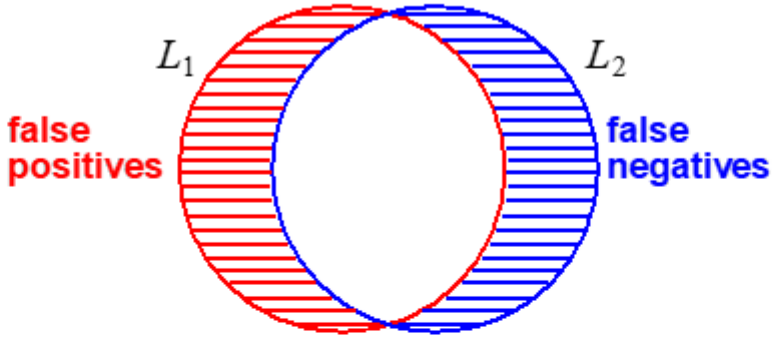
- Noun → stench | breeze | glitter | nothing | Wumpus | pit | pits | gold | east | …
- Verb → is | see | smell | shoot | feel | stinks | go | grab | carry | kill | turn | …
- Adjective → right | left | east | south | back | smelly | …
- Adverb → here | there | nearby | ahead | right | left | east | south | back | …
- Pronoun → me | you | I | it | …
- Name → John | Mary | Boston | UCB | PAJC | …
- Article → the | a | an | …
- Preposition → to | in | on | near | …
- Conjunction → and | or | but | …
- Digit → 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

Wumpus Grammar

S	→	NP VP	I + feel a breeze
	|	S Conjunction S	I feel a breeze + and + I smell a wumpus
NP	→	Pronoun	I
	|	Noun	pits
	|	Article Noun	the + wumpus
	|	Digit Digit	3 4
	|	NP PP	the wumpus + to the east
	|	NP RelClause	the wumpus + that is smelly
VP	→	Verb	stinks
	|	VP NP	feel + a breeze
	|	VP Adjective	is + smelly
	|	VP PP	turn + to the east
	|	VP Adverb	go + ahead
PP	→	Preposition NP	to + the east
RelClause	→	that VP	that + is smelly

Grammaticality Judgements

Formal language L_1 may differ from natural language L_2



Adjusting L_1 to agree with L_2 is a learning problem!

* the gold grab the wumpus
* I smell the wumpus the gold
  I give the wumpus the gold
* I donate the wumpus the gold

Intersubjective agreement somewhat reliable, independent of semantics!
Real grammars 10–500 pages, insufficient even for "proper" English

Parse Trees

Exhibit the grammatical structure of a sentence

I shoot the wumpus

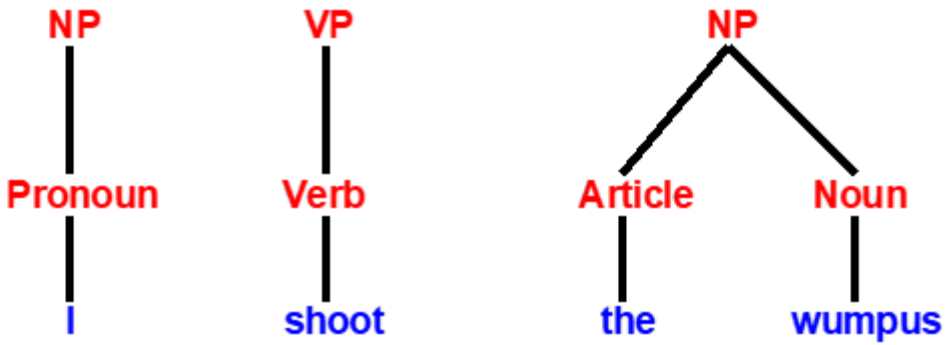
Parse Trees

Exhibit the grammatical structure of a sentence



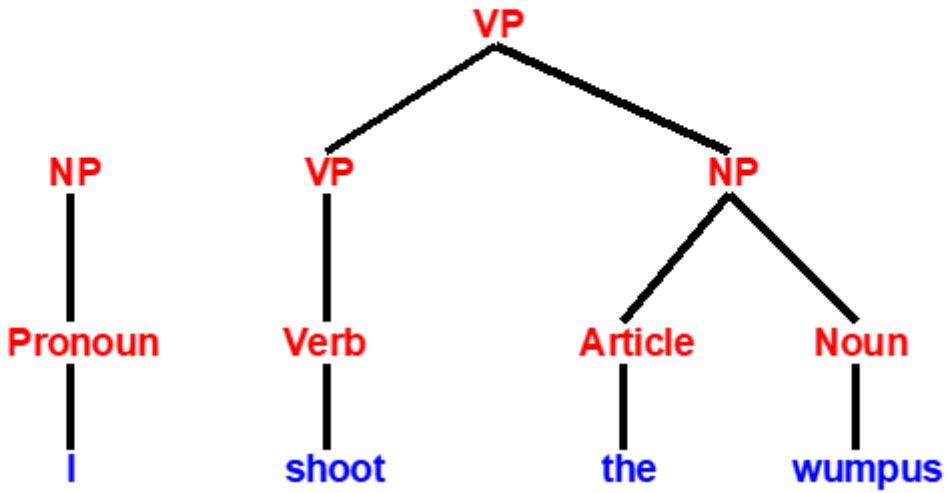
Parse Trees

Exhibit the grammatical structure of a sentence



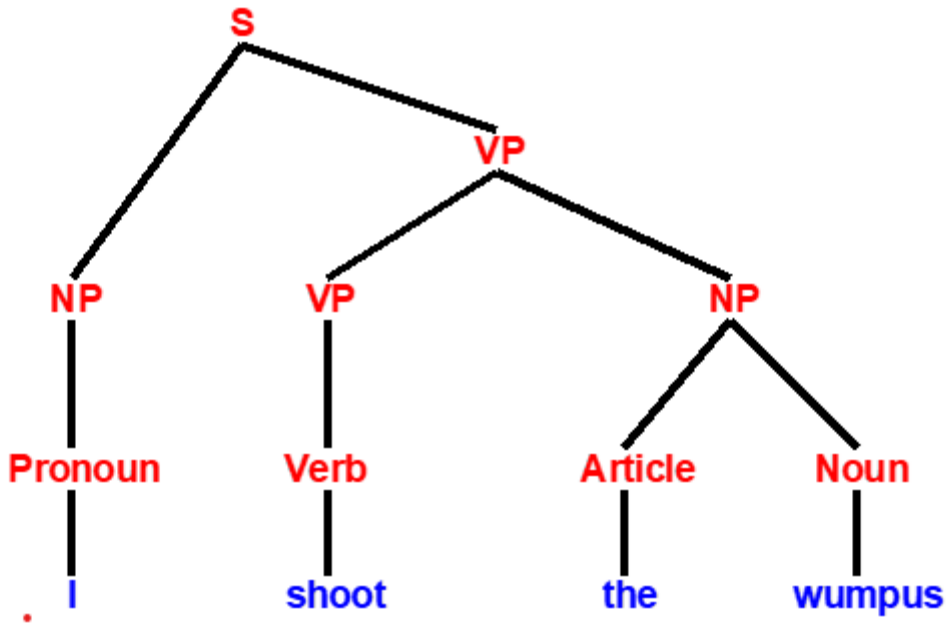
Parse Trees

Exhibit the grammatical structure of a sentence



Parse Trees

Exhibit the grammatical structure of a sentence



Syntax in NLP

- Most view syntactic structure as an essential step towards meaning;
  - "Mary hit John" ≠ "John hit Mary"
  - "And since I was not informed – as a matter of fact, since I did not know that there were excess funds until we, ourselves, in that checkup after the whole thing blew up, and that was, if you'll remember, that was the incident in which the attorney general came to me and told me that he had seen a memo that indicated there were no more funds"

Syntax in NLP

- Most view syntactic structure as an essential step towards meaning;
  - "Mary hit John" ≠ "John hit Mary"
  - "And since I was not informed – as a matter of fact, since I did not know that there were excess funds until we, ourselves, in that checkup after the whole thing blew up, and that was, if you'll remember, that was the incident in which the attorney general came to me and told me that he had seen a memo that indicated there were no more funds"
  - "Wouldn't the sentence 'I want to put a hyphen between the words Fish and And and And and Chips in my Fish-And-Chips sign' have been clearer if quotation marks had been placed before Fish, and between Fish and and, and and and And, and And and and, and and and And, and And and and, and and and Chips, as well as after Chips?

Real Language

- Real human languages provide many problems for NLP:
  - Ambiguity
  - Anaphora
  - Indexicality
  - Vagueness
  - Discourse structure
  - Metonymy
  - Metaphor
  - Noncompositionality

Ambiguity

- Squad helps dog bite victim

Ambiguity

- Squad helps dog bite victim
- Helicopter powered by human flies

Ambiguity

- Squad helps dog bite victim
- Helicopter powered by human flies
- I ate spaghetti with meatballs

Ambiguity

- Squad helps dog bite victim
- Helicopter powered by human flies
- I ate spaghetti with meatballs

  salad

Ambiguity

- Squad helps dog bite victim
- Helicopter powered by human flies
- I ate spaghetti with meatballs

  salad

  abandon

Ambiguity

- Squad helps dog bite victim
- Helicopter powered by human flies
- I ate spaghetti with meatballs

  salad

  abandon

  a fork

Ambiguity

- Squad helps dog bite victim
- Helicopter powered by human flies
- I ate spaghetti with meatballs

  salad

  abandon

  a fork

  a friend

Ambiguity

- Squad helps dog bite victim
- Helicopter powered by human flies
- I ate spaghetti with meatballs

  salad

  abandon

  a fork

  a friend

Ambiguity can be lexical (polysemy), syntactic, semantic, referential

Anaphora

- Using pronouns to refer back to entities already introduced in the text
- After Mary proposed to John, they found a preacher and got married.

Anaphora

- Using pronouns to refer back to entities already introduced in the text
- After Mary proposed to John, they found a preacher and got married.
- For the honeymoon, they went to Hawaii.

Anaphora

- Using pronouns to refer back to entities already introduced in the text
- After Mary proposed to John, they found a preacher and got married.
- For the honeymoon, they went to Hawaii.
- Mary saw a ring through the window and asked John for it.

Anaphora

- Using pronouns to refer back to entities already introduced in the text
- After Mary proposed to John, they found a preacher and got married.
- For the honeymoon, they went to Hawaii.
- Mary saw a ring through the window and asked John for it.
- Mary threw a rock at the window and broke it.

Anaphora

- Using pronouns to refer back to entities already introduced in the text
- After Mary proposed to John, they found a preacher and got married.
- For the honeymoon, they went to Hawaii.
- Mary saw a ring through the window and asked John for it.
- Mary threw a rock at the window and broke it.
- *Compare*: Mary threw a vase at the wall and broke it.

Indexicality

- Indexical sentences refer to utterance situation (place, time, etc)
- I am over here.
- Why did you do that?

Metonymy

- Using one noun phrase to stand for another
- I've read Shakespeare
- Chrysler announced record profits
- The ham sandwich on Table 4 wants another beer.

Metaphor

- “Non-literal” usage of words and phrases, often systematic:
- I’ve tried killing the process, but it won’t die. Its parent keeps it alive.

Noncompositionality

- Meaning of a phrase = combined meaning of its terms?

Noncompositionality

- Basketball shoes

Noncompositionality

- Basketball shoes
- Baby shoes

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes
- Brake shoes

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes
- Brake shoes

- Red book

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes
- Brake shoes

- Red book
- Red pen

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes
- Brake shoes

- Red book
- Red pen
- Red hair

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes
- Brake shoes

- Red book
- Red pen
- Red hair
- Red herring

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes
- Brake shoes

- Red book
- Red pen
- Red hair
- Red herring

- Small moon

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes
- Brake shoes

- Red book
- Red pen
- Red hair
- Red herring

- Small moon
- Large molecule

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes
- Brake shoes

- Red book
- Red pen
- Red hair
- Red herring

- Small moon
- Large molecule
- Mere child

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes
- Brake shoes

- Red book
- Red pen
- Red hair
- Red herring

- Small moon
- Large molecule
- Mere child
- Alleged murderer

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes
- Brake shoes

- Red book
- Red pen
- Red hair
- Red herring

- Small moon
- Large molecule
- Mere child
- Alleged murderer
- Real leather

Noncompositionality

- Basketball shoes
- Baby shoes
- Alligator shoes
- Designer shoes
- Brake shoes

- Red book
- Red pen
- Red hair
- Red herring

- Small moon
- Large molecule
- Mere child
- Alleged murderer
- Real leather
- Artificial grass

NLP in the Real World



Jeremy Fowler 
@JFowlerESPN

Follow 

I'm not expecting the Steelers to make a move for cornerback help at the deadline barring unforeseen development. Will likely roll with what they have.

12:19 PM - 30 Oct 2018



Steeler 4 Life @jarvis1104 · Oct 30

Replying to @JFowlerESPN @Alex_Kozora

Sadly this tweet can be sent every year at this time and remain true

7 80



John @jrw412 · Oct 30

Replying to @JFowlerESPN

Huge surprise

1 12



jONh @jonhthagreat · Oct 30

not really, as much as i love the steelers they are probably the most boring team when it comes to making deadline deals

1 1 6



my name jason @jbmorton21 · Oct 30

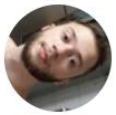
You're not that great at identifying sarcasm

1 9



jONh @jonhthagreat · Oct 30

lol i didnt even think of sarcasm for some reason 😂



Daniel Cabral Nephas @medanielrose · Oct 30

Replying to @JFowlerESPN @SteelTownUsa

Dude, dont u know words can hurt? U dont have to say it just cuz its true



@BurghSportsRT · Oct 30

Replying to @JFowlerESPN

With all that money they got from bell being out .. didn't take the opportunity of what was given



Michael Perrotti @mperrotti22 · Oct 30

Replying to @JFowlerESPN



NLP in the Real World



NLP in the Real World

- Thinking beyond agents, how you do NLP depends on what task you're trying to accomplish
- Examples
  - Part-of-speech classification (noun/verb/etc.)
  - Classification w.r.t. a taxonomy (e.g., Amazon products)
  - Authorship detection (e.g., Founding Fathers)
  - Language detection (which language is this?)
  - Sentiment analysis (is this positive, negative, other?)
  - Concept extraction, summarization (what is this about?)
  - Machine translation (one language to another)
  - Language generation (e.g., Chat-GPT)

NLP in Python

- Let's check out a robust Python library for doing NLP work: spaCy (plus spacytextblob)

```
> python -m pip install spacy spacytextblob
```

- According to their website…
  - spaCy is a free, open-source library for advanced NLP in Python
  - spaCy is **not** a platform, an API, an out-of-the-box chatbot engine, or research software
  - (though they've recently added integration with chatbots, LLMs, etc.)
  - For more info, see https://spacy.io/usage/spacy-101

nlp.py

```
import spacy
from spacytextblob.spacytextblob import SpacyTextBlob

# before running, you'll need the spacy English model and the textblob corpora:
# > python -m spacy download en_core_web_sm
# > python -m textblob.download_corpora

# load the English model
nlp = spacy.load('en_core_web_sm')

# define the text to analyze
text = 'Apple is looking at buying U.K. startup for $1 billion'
print(f'\nText: "{text}"')

# analyze the text to create a spacy augmented "document"
doc = nlp(text)

# print each token along with some of its properties
print(f'\nToken    Lemma    POS    Shape   Alpha  Stop\n' + ('-' * 49))
for token in doc:
    print(f'{token.text:<9} {token.lemma_:<9} {token.pos_:<7} ' +
          f'{token.shape_:<7} {token.is_alpha:<6} {token.is_stop:<6}')

# print the named entities
print(f'\nEntity      Start   End    Label\n' + ('-' * 36))
for ent in doc.ents:
    print(f'{ent.text:<12} {ent.start_char:<7} {ent.end_char:<7} {ent.label_:<8}')
```

```
> python nlp.py

Text: "Apple is looking at buying U.K. startup for $1 billion"

Token    Lemma    POS    Shape   Alpha  Stop
-----------------------------------------------
Apple    Apple    PROPN  Xxxxx   1      0
is       be       AUX    xx      1      1
looking  look     VERB   xxxx    1      0
at       at       ADP    xx      1      1
buying   buy      VERB   xxxx    1      0
U.K.     U.K.     PROPN  X.X.    0      0
startup  startup  NOUN   xxxx    1      0
for      for      ADP    xxx     1      1
$        $        SYM    $       0      0
1        1        NUM    d       0      0
billion  billion  NUM    xxxx    1      0

Entity      Start  End    Label
---------------------------------
Apple       0      5      ORG
U.K.        27     31     GPE
$1 billion  44     54     MONEY
```

Token Information

Token: The original word text

Lemma: The base form of the word

POS: The part-of-speech tag

Shape: The word shape – capitalization, punctuation, digits

Alpha: Is the token alphabetic?

Stop: Is the token a "stop word"?

Named Entities

ORG = Organization

GPE = Geopolitical Organization

MONEY = Currency

nlp.py

```
...

# add spacytextblob to pipeline for sentiment analysis
nlp.add_pipe('spacytextblob')

# run sentiment analysis on sample texts

def analyze_sentiment(text):
    doc = nlp(text)
    print(f'\nText:          "{text}"')
    print(f'Polarity:      {doc._.blob.polarity:.2f}')
    print(f'Subjectivity:   {doc._.blob.subjectivity:.2f}')

analyze_sentiment("It's great that Apple is buying a U.K. startup")
analyze_sentiment("It's not cool that Apple is buying a U.K. startup")
analyze_sentiment("It's ridiculous that Apple is buying a U.K. startup")
```

Text: "It's great that Apple is buying a U.K. startup"
Polarity: 0.80
Subjectivity: 0.75

Text: "It's not cool that Apple is buying a U.K. startup"
Polarity: -0.17
Subjectivity: 0.65

Text: "It's ridiculous that Apple is buying a U.K. startup"
Polarity: -0.33
Subjectivity: 1.00

Polarity – ranges from -1 to 1,
where 1 is positive sentiment,
-1 is negative sentiment,
and 0 is neutral sentiment

Subjectivity — ranges from 0 to 1
where 0 is objective
and 1 is subjective)

NLP in the Real World

- Problem: A perfect, unambiguous parsed understanding of a sentence is often elusive.
- In the face of real-world language & ambiguity, how can we handle the explosion of possible grammatical rules?
  - Allow for partially-formed parse trees
  - Incorporate probabilistic parsing
  - Use modern techniques (e.g., neural networks) that can better handle “imperfect” language

Coming up…

- These ideas lead us naturally into the next topic, Machine Learning
  - How can agents learn what to do and when to do it?
  - Once they know what to do, how can they improve their behavior over time?